Computer Networks

Computing and Mathematics

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Computer Networks (A11144)

Short Title:	Computer Networks		
Faculty	Science and Computing		
Department	Computing and Mathematics		
Cluster	Networks and Cloud		
Author	EREADE		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module introduces Computer Networking terminology, network protocols and models. Students will use simulation and protocol analysis software to configure network devices and explore various network protocol operations. A detailed examination of TCP/IP, IP addressing and Ethernet is presented. A brief introduction to Routing, Network management and Wireless LANs is also provided. Practical skills are an essential part of this module.

Programmes

`COMP-0606`	BSc (Hons) in Applied Computing (International) (`WD_KACMI_B`)	2 / 3 / M
`COMP-0637`	BSc (Hons) in Applied Computing (`WD_KACCM_B`)	2 / 3 / M
`COMP-0637`	BSc (Hons) in Applied Computing (`WD_KCOMP_B`)	2 / 3 / M
`COMP-0637`	BSc (Hons) in Computer Forensics and Security (`WD_KCOFO_B`)	2 / 3 / M
`COMP-0637`	BSc (Hons) in Computer Science (`WD_KCMSC_B`)	2 / 3 / M
`COMP-0637`	BSc (Hons) in Software Engineering (`WD_KDEVP_BI`)	2 / 3 / M
`COMP-0637`	BSc (Hons) in Software Systems Development (`WD_KDEVP_B`)	2 / 3 / M
`COMP-0637`	BSc (Hons) in the Internet of Things (International) (`WD_KINTT_BI`)	2 / 3 / M
`COMP-0637`	BSc in Applied Computing (`WD_KCOMP_D`)	2 / 3 / M
`COMP-0637`	BSc in Information Technology (`WD_KINFT_D`)	2 / 3 / M
`COMP-0637`	BSc in Software Systems Development (`WD_KCOMC_D`)	2 / 3 / M

Indicative Content

- Introduction to Computer Networks and Protocols
- OSI and TCP/IP models
- Ethernet and VLANs
- IPv4 Addressing and subnetting
- IPv6
- Routing
- Transport Layer Protocols and Functionality
- Application Layer Protocols and Functionality e.g. HTTP, FTP, DNS, SMTP
- Wireless LANs
- Network Management

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Use network protocol models and tools to explain communications in data networks.
2. Describe in detail the major components, operation and functionality of a computer network and commonly used protocols and services.
3. Construct an IPv4/IPv6 addressing design solution.
4. Build a simple network using routers and switches.
5. Use Cisco command line interface to perform basic router and switch configuration.
6. Implement a basic wireless network.
7. Describe basic computer network management concepts.

Learning and Teaching Methods

- The practical lab component will be delivered in one double lab session.
- There is a strong emphasis on practical, lab-based exercises.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Practical	60%	3,4,5,6
In-Class Assessment	40%	1,2,3,4,5,6,7

Assessment Criteria

<40%: Unable to describe the major functions and operation of a Computer Network. Unable to describe and compare the OSI and TCP/IP models. Poor understanding of the role of communications protocols in computer networks.

40%-49%: Can describe and compare the OSI and TCP/IP models. Can provide overview of main computer network components and protocols.

50%-59%: All of the above. Can describe in detail the data encapsulation process. Demonstrate an understanding of basic LAN implementation.

60%-69%: In addition, be able to recommend a network solution given an organisation's requirements.

70%-100%: All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of both complex and unforeseen problems through the use and modification of appropriate skills and tools.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	12
Practical	24	12
Independent Learning	75	111

Essential Material

- "Cisco Network Academy." https://www.netacad.com/

Supplementary Material

- Cisco, Networking. _Network Basics, CCNA Routing & Switching Companion Guide_. NY: Cisco Press, 2014.
- Tanenbaum, A. and D. Wetherall. _Computer Networks_. 5th Ed. New York: Pearson Education, 2013.

Requested Resources

- COMPUTER LAB: Networks Lab